

ASSIGNMENT – REGRESSION ALGORITHM

1. Identify your problem statement.

Predict the insurance charges based on the input independent fields - age, sex, bmi, children, smoker.

2. Tell basic info about the dataset (Total number of rows, columns).

To predict the insurance charges, we have dataset with five independent fields (age, sex, bmi, children, smoker) and one dependent field (charges).

3. Mention the pre-processing method if you're doing any (like converting string to number – nominal data)

Independent sex (values - male, female), smoker (values - yes, no) needs to convert into number using one hot encoding

4. Develop a good model with r2_score. You can use any machine learning algorithm; you can create many models. Finally, you have to come up with final model.

4a. MULTIPLE LINEAR REGRESSION (R^2 value: 0.7894790)

4b. SUPPORT VECTOR MACHINE (SVM)

S.No	Kernel	C	R2 value
1	Linear	10	0.46593
2	Linear	100	0.63124
3	Linear	500	0.764481
4	Linear	1000	0.767145
5	Linear	2000	0.747843
6	Linear	3000	0.745374
7	poly	10	0.039066
8	poly	100	0.615655
9	poly	500	0.823464
10	poly	1000	0.852055
11	poly	2000	0.856237
12	poly	3000	0.855319
13	rbf	10	-0.03316
14	rbf	100	0.31379
15	rbf	500	0.661926
16	rbf	1000	0.811494
17	rbf	2000	0.85833
18	rbf	3000	0.869845
19	sigmoid	10	0.040195
20	sigmoid	100	0.531425
21	sigmoid	500	0.444274
22	sigmoid	1000	0.287398
23	sigmoid	2000	-0.58788
24	sigmoid	3000	-2.09153

4c. DECISION TREE(DT) (R^2 value without hyper tuning: 0.68021)

S.No	CRITERION	MAX_FEATURES	SPLITTER	R2 VALUE
1	squared_error	None	best	0.687201
2	squared_error	None	random	0.705952
3	squared_error	sqrt	best	0.719819
4	squared_error	sqrt	random	0.637043
5	squared_error	log2	best	0.73046
6	squared_error	log2	random	0.604122
7	absolute_error	None	best	0.666275
8	absolute_error	None	random	0.72457
9	absolute_error	sqrt	best	0.77394
10	absolute_error	sqrt	random	0.691147
11	absolute_error	log2	best	0.706193
12	absolute_error	log2	random	0.632678
13	poisson	None	best	0.726989
14	poisson	None	random	0.701526
15	poisson	sqrt	best	0.74164
16	poisson	sqrt	random	0.70546
17	poisson	log2	best	0.662986
18	poisson	log2	random	0.680353

4d. RANDOM FOREST (RF) (R^2 value = 0.8529170)

S.No	n_estimators	random_state	criterion	R2 value
1	50	0	squared_error	0.84983
2	50	0	absolute_error	0.85266
3	50	0	poisson	0.8491
4	100	0	squared_error	0.852009
5	100	0	absolute_error	0.85383
6	100	0	poisson	0.852633

5. All the research values (r2_score of the models) should be documented. (You can make tabulation or screenshot of the results.)

Highest R^2 value is 0.869845 using SVM with hyper tuning parameter. Below is the screenshot attached.

Preview

Code

Blame

773 lines (773 loc) · 21.3 KB



```
In [9]: from sklearn.svm import SVR
        regressor = SVR(kernel='rbf', C=3000)
        regressor = regressor.fit(X_train, y_train)
```

```
C:\Anaconda\Lib\site-packages\sklearn\utils\validation.py:1406: DataConversionWarning:
d. Please change the shape of y to (n_samples, ), for example using ravel().
y = column_or_1d(y, warn=True)
```

```
In [10]: y_prediction = regressor.predict(X_test)
```

```
In [11]: from sklearn.metrics import r2_score
        r_square_score = r2_score(y_test, y_prediction)
        r_square_score
```

```
Out[11]: 0.8698453120308981
```

```
In [12]: import pickle
        filename = "finalized_model_SVM.sav"
        pickle.dump(regressor, open(filename, 'wb'))
```

```
In [13]: loaded_model = pickle.load(open(filename, 'rb'))
        result = loaded_model.predict([[1234, 345, 4565, 1, 0]])
        result
```

```
Out[13]: array([16589.18908958])
```

```
In [ ]:
```

6. Mention your final model, justify why u have chosen the same

My final model is SVM using hyper tuning. Since other model R^2 value is less than SVM using hyper tuning model. Please find the code GIT link.

GIT link: [machineLearning/Week-3-ML-Regression/Assignments/SVM_Assignment](https://github.com/ababu3010/machineLearning/Week-3-ML-Regression/Assignments/SVM_Assignment) at main · ababu3010/machineLearning